COURSE STRUCTURE

Course Code	**CSE20060**				
Course Category	**Program Major**				
Course Title	**Database Management System**				
Teaching Scheme	**Lectures**	**Tutorials**	**Laboratory / Practical**	**Project**	**Total**
Weekly load hours	**3**	--	--	--	**3**
Credits	**3**	--	--	--	**3**
Assessment Schema Code	**TT1**				

Prerequisites:

- Data Structures
- Discrete Structures

Course Objectives:

1. Knowledge:

i. Understand the fundamental concepts of database management System
ii. To provide a strong formal foundation in database concepts, DBMS architectures recent technologies and best industry practices.

2. Skill:

i. To learn the SQL database system.
ii. To program PL/SQL including stored procedures, stored functions, cursors and packages.

3. Attitude

i. To design the database system for real world applications.
ii. To access, modify, program and authenticate the database system.

Course Outcomes:

After completion of this course students will be able to:

1. Understand the different data models.
2. Implement relational database design from any data model.
3. Create database system for real world applications
4. Handle the transaction management system.

Course Contents:

Unit I - Introduction to DBMS and Data Modelling

DBMS Vs File Systems, Database System Architecture, Database Architectures: Centralized, Client-Server, Parallel, Distributed, Data Abstraction, Data Independence, Data Definition and Data Manipulation Languages, Data Models, E-R diagram: Components of E-R Model, Conventions, Keys, EER diagram Components, Reduce E-R diagram into Tables

Associate Dean
Academics
(School)

Dean Academics
MITWPU

Registrar
MITWPU

MIT-WPU, PUNE-38.
APPROVED BY
04 JUN 2025
ACADEMIC COUNCIL

Unit II - Relational Database Design and Normalisation

Relational Model: Attributes, Tuple, Domain, CODD's rule Relational Integrity, Referential Integrities, Enterprise Constraints, Normalization: 1NF, 2NF, 3NF, BCNF, Functional dependency, Decomposition Query Processing: Overview, Measures of Query cost, Selection and Join operations, Evaluation of Expressions.

Unit III - Relational Algebra and Database Programming

Relational Algebra, Basic Operations, Relational calculus: Tuple Calculus, Domain Calculus, Introduction to SQL, Characteristics and advantages of SQL, SQL Data Types, DDL Commands, DCL Commands. SQL Queries: DML Queries with Select Query Clauses, Creating, Modifying, Deleting.

Views: Creating, Dropping, Updating, Indexes, SQL DML Queries, Set Operations, Predicates and Joins, set membership, Grouping and Aggregation, Aggregate Functions, Nested Queries

Unit IV - Introduction to Programming in SQL(PL/SQL)

PL/SQL Concepts: Basic PL/SQL block, PL/SQL Functions and Procedures, Different modes in procedure, Difference between functions and procedures, Cursors, Database Triggers. Exception Handling in PL/SQL. Application of DBMS

Unit V - Transaction Management and Concurrency Control

ACID properties, transactions, schedules and concurrent execution of transactions, serializability: View, Conflict. Concurrency control lock-based protocol (simple, 2 phases: Rigorous 2 phase, Strict 2 phase), Cascade-less Schedule, Recoverable Schedule,

Deadlocks: Prevention Techniques (Wait Die, Wound Wait), Detection Techniques, Database Recovery: Failure classification Recovery and atomicity: Log-based recovery, Shadow paging.

Learning Resources:

Text Books:

1. Abraham Silberschatz, Henry F. Korth and S. Sudarshan, Database System Concepts 6th Ed, McGraw Hill, 2010.
2. Elmasri R. and Navathe S.B., "Fundamentals of Database Systems", 4th Edition, Pearson Education.

Reference Books:

3. Ramakrishnan, R. and Gherke, J., "Database Management Systems", 3rd Ed., McGraw-Hill.
4. Connolly T, Begg C.,Database Systems: A Practical Approach to Design, Implementation, and Management, 6th Edition

Laboratory Reference Books:

5. Ivan Bayross, "SQL, PL/SQL: The Programming Language of Oracle", 3rd Edition BPB Publication
6. Reese G., Yarger R., King T., Williums H, "Managing and Using MySQL", O'Reilly Media, Inc., ISBN: 9780596002114, 2nd Edition

Web Resources:

7. https://www.mysql.com/
8. https://dev.mysql.com/doc/refman/8.0/en/create-procedure.html
9. https://www.oreilly.com/library/view/managing-using/0596002114/

Associate Dean
Academics
(School)

Dean Academics
MITWPU

Registrar
MITWPU

MIT-WPU, PUNE-38.
APPROVED BY
04 JUN 2025
ACADEMIC COUNCIL